ANSWER KEY & MARKING SCHEME SET 1

Chapter 9: **Light - Reflection and Refraction** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** straight lines	Rectilinear propagation.
2	**(c)** refraction	Definition.
3	**(b)** two spherical surfaces	Basic definition.
4	**(b)** reflection	Definition.
5	**(b)** towards the normal	Water is denser.
6	**(b)** centre	Bulges outwards.
7	**(b)** the angle of reflection	Law of reflection.
8	**(b)** vacuum	3×10^8 m/s.
9	**(b)** edges	Thin at the centre.
10	**(b)** normal	It is the normal.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Light bounces back from a polished surface (1); Surface of the mirror (1).	1+1
12	Convex mirror (1); Virtual and erect (1).	1+1
13	Glass (1.5) (1); Water is 1.33 (1).	1+1
14	Virtual and erect (1); Same size as the object (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Shaving: concave (1); Torch: concave (1); Rear-view: convex (1).	1+1+1
16	n = c/v (1); n = 3×10^8 / 1.25×10^8 (1); n = 2.4 (1).	1+1+1
17	1/v - 1/u = 1/f (1); m = v/u (1); m = h'/h (1).	1+1+1
	OR: Angle of incidence = angle of reflection (1); Incident ray, reflected ray and normal in the same plane (1); Both apply to all reflecting surfaces (1).	1+1+1
18	Speed of light changes in water (1); Light slows down in a denser medium (1); It bends towards the normal (1).	1+1+1
	OR: A straight line through the pole and centre of curvature (1); Normal to the mirror at the pole (1); Also passes through the principal focus (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	At infinity: image at F, real, point (1); At C: image at C (1); At C: real, inverted, same size (1); Between P and F: behind the mirror (1); Virtual, erect, enlarged (1).	1+1+1+1+1
	OR: 1/v = -1/20 + 1/40 (1); 1/v = -1/40 (1); v = -40 cm (1); Image at C (1); Real, inverted, same size (1).	1+1+1+1+1
20	Image is always erect (1); Image is diminished (1); Field of view is wide (1); Driver sees more traffic (1); Image is virtual and behind the mirror (1).	1+1+1+1+1
	OR: 1/v = 1/10 - 1/20 (1); 1/v = 1/20 (1); v = 20 cm (1); m = v/u = -1 (1); Real, inverted, same size (1).	1+1+1+1+1

Mistakes to avoid:

- Ignoring the sign convention: for a concave mirror or lens use $u < 0$ and, for a concave mirror, $f < 0$ (convex lens $f > 0$, concave lens $f < 0$).
- Using the lens formula $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ for mirrors, or the mirror formula $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$ for lenses.
- Using f in centimetres while calculating power; $P = 1/f$ gives dioptres only when f is in **metres**.
- Saying that light bends away from the normal when entering glass; it bends towards the normal in a denser medium.
- Describing the image formed by a concave mirror when the object is between P and F as real; it is virtual, erect and enlarged.
- Writing the refractive index as $\sin r / \sin i$; Snell's law gives $n = \sin i / \sin r$ for light going from air into the medium.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 9: **Light - Reflection and Refraction** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 3×10^8 m/s	Standard value.
2	**(b)** optical centre	Optical centre O.
3	**(c)** virtual	Cannot be taken on a screen.
4	**(b)** 1	n = c/c.
5	**(d)** dioptre	D.
6	**(b)** converging mirror	It converges parallel rays.
7	**(b)** no unit	It is a ratio.
8	**(b)** positive	f is positive.
9	**(d)** diverging mirror	It diverges parallel rays.
10	**(c)** optically denser	n is larger.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	f = R/2 (2).	2
12	P = 1/f (1); f in metres (1).	1+1
13	Centre of the reflecting surface (1); Lies on the mirror (1).	1+1
14	Bending of light on passing from one medium to another (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Virtual (1); Erect and same size (1); Laterally inverted (1).	1+1+1
16	P = 1/f (1); P = 1/0.2 (1); P = +5 D if convex and -5 D if concave (1).	1+1+1
17	Magnifying glass (1); Camera (1); Spectacles for hypermetropia (1).	1+1+1
	OR: R = 2f (1); R = 2 x 10 (1); R = 20 cm (1).	1+1+1
18	f = 1/P (1); f = 1/2 m (1); f = 50 cm (1).	1+1+1
	OR: Real image forms on a screen (1); Virtual cannot be formed (1); Virtual images are seen only by looking into the mirror or lens (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Incident ray, refracted ray and normal in same plane (1); sin i / sin r = constant (1); Constant is the refractive index (1); Valid for a given colour (1); Ray bends towards normal in denser medium (1).	1+1+1+1+1
	OR: Objects are placed to the left of the mirror (1); Distances measured from the pole (1); Right of the pole: positive (1); Left of the pole: negative (1); Upward heights positive, downward negative (1).	1+1+1+1+1
20	n = c/v (1); v = c/n (1); v = 3×10^8 / 1.33 (1); v = 2.26×10^8 m/s (1); Light is slower in water (1).	1+1+1+1+1
	OR: Power is reciprocal of the focal length in metres (1); Unit dioptre (1); f = -25 cm gives -4 D (1); f = +50 cm gives +2 D (1); Convex lenses have positive power (1).	1+1+1+1+1

Mistakes to avoid:

- Ignoring the sign convention: for a concave mirror or lens use $u < 0$ and, for a concave mirror, $f < 0$ (convex lens $f > 0$, concave lens $f < 0$).
- Using the lens formula $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ for mirrors, or the mirror formula $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$ for lenses.
- Using f in centimetres while calculating power; $P = 1/f$ gives dioptres only when f is in **metres**.
- Saying that light bends away from the normal when entering glass; it bends towards the normal in a denser medium.
- Describing the image formed by a concave mirror when the object is between P and F as real; it is virtual, erect and enlarged.
- Writing the refractive index as $\sin r / \sin i$; Snell's law gives $n = \sin i / \sin r$ for light going from air into the medium.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 9: **Light - Reflection and Refraction** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** negative	Sign convention.
2	**(c)** concave	It sends a parallel beam.
3	**(b)** refraction	Rays bend at the surface.
4	**(c)** +1 D	1/1 m.
5	**(d)** focal length	f.
6	**(a)** 1.5	Common glass.
7	**(d)** image size / object size	Definition of m.
8	**(a)** twice the focal length	R = 2f.
9	**(a)** glass to air	Denser to rarer.
10	**(c)** convex lens	Used to see enlarged images.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Converging lens (1); Thicker at the centre (1).	1+1
12	Radius of the sphere of which the mirror is a part (2).	2
13	Ratio of speed of light in vacuum to that in a medium (2).	2
14	Convex: positive (1); Concave: negative (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	f = R/2 (1); f = 30/2 (1); f = 15 cm (1).	1+1+1
16	Image forms at the focus (1); Real and inverted (1); Highly diminished (1).	1+1+1
17	Perpendicular distance between incident and emergent rays (1); Yes, emergent ray is parallel (1); It happens in a glass slab (1).	1+1+1
	OR: Ray parallel to axis passes through F after reflection (1); Ray through F becomes parallel to the axis (1); Ray through C returns along the same path (1).	1+1+1
18	Convex is thicker at the centre; concave thinner (1); Convex converges; concave diverges (1); Convex has positive f; concave has negative f (1).	1+1+1
	OR: It is negative (1); The lens is divergent (1); Parallel rays appear to diverge from the focus (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Ray bends towards normal at entry (1); It bends away at exit (1); Emergent ray is parallel to the incident ray (1); It shows a lateral displacement (1); Displacement depends on thickness and angle (1).	1+1+1+1+1
	OR: Concave: shaving mirror, headlights (1); Concave: used by dentists (1); Convex: rear-view mirror (1); Convex: road safety at turns (1); Convex gives a wide view (1).	1+1+1+1+1
20	At 2F: image at 2F on the other side (1); Real, inverted, same size (1); Between F and O: virtual, erect, enlarged (1); At infinity: image at F (1); Real, inverted, point-sized (1).	1+1+1+1+1
	OR: A ray parallel to the axis diverges (1); It appears to come from F (1); Ray through O goes straight (1); The image is between O and F (1); Virtual, erect and diminished (1).	1+1+1+1+1

Mistakes to avoid:

- Ignoring the sign convention: for a concave mirror or lens use $u < 0$ and, for a concave mirror, $f < 0$ (convex lens $f > 0$, concave lens $f < 0$).
- Using the lens formula $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ for mirrors, or the mirror formula $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$ for lenses.
- Using f in centimetres while calculating power; $P = 1/f$ gives dioptres only when f is in **metres**.

- Saying that light bends away from the normal when entering glass; it bends towards the normal in a denser medium.
- Describing the image formed by a concave mirror when the object is between P and F as real; it is virtual, erect and enlarged.
- Writing the refractive index as $\sin r / \sin i$; Snell's law gives $n = \sin i / \sin r$ for light going from air into the medium.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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